Restrictions of the Alcatel Microelectronics CMOS 0.35 µm technology on Europractice MPW runs.

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On EUROPRACTICE MPW runs in CMOS 0.35 µm following restrictions apply :

1. EUROPRACTICE MPW service uses the **C035M-A** for mixed analog-digital technology and **C035M-D** technology for pure digital circuits. The process is a **FIVE METAL LAYER** process.
2. Layers
   - C035M-D : Use the layers as specified on page 17 of DS13330 paragraph 4.6 for pure digital circuits.
   - C035M-A : Use the layers as specified on page 9 of DS13337 for mixed analog-digital circuits.
   - General requirements for IGS numbering (page13 of DS13330).
3. Read carefully the dummy metal generation rule on page 14 paragraph 4.4.4 of DS13330.

   For analog circuits, the designer must cover the sensitive data by IGS layer 63 (metal_disable) to avoid dummy metal generation on those areas. In case the customer wants to prototype a circuit that is likely to go into volume production later on, please contact IMEC for metal area coverage.
4. Read carefully the general layout requirements on page 25 of DS13330.
   - Product identification (6.3 on page 26 of DS13315) has not to be done. This is done by Europractice.
   - **!!! VERY IMPORTANT page 25 of DS13330 !!!**

   The layout grid size has to be minimum 0.05 µm for all layers. The layers poly and contact must use the 0.025 µm. The edges of all polygons must be on grid.
5. Read carefully the **plasma induced charging rules** 9.5 on page 88 of DS13330.
6. Probe pads [(M4, M5, Via4, Overlay, Nwell) or (M5, Overlay, Nwell) etc.] may be used with a minimum area of 15x15 µm for overlay.
7. Scribe lanes have not to be added (page 77 of DS13330). This is done by EUROPRACTICE.
8. EUROPRACTICE will use document "EUROPRACTICE ASSEMBLY RULES" instead of Spec 13600 'ASSEMBLY LAY-OUT RULES' which is mentioned in doc DS13337 on page 4 and page 6 of DS13330. (Available on web-site: ASIC prototyping → Chip assembly)